ORIGINAL INVESTIGATIONS

A Randomized, Double-Blind, Placebo-Controlled Trial to Evaluate the Effect of Enalapril in Patients With Clinical Diabetic Nephropathy

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● It is unknown if the antiproteinuric effect of angiotensin-converting enzyme (ACE) inhibitors reflects attenuation in the rate of progression of diabetic nephropathy. We report the results of a randomized, double-blind clinical trial designed to evaluate the longitudinal (18-month) effect of the ACE inhibitor, enalapril (5 to 40 mg/d), versus a placebo on 24-hour urinary protein excretion and on the rate of progression of renal disease in 33 patients with clinical diabetic nephropathy. Systemic blood pressure was controlled throughout the trial with conventional antihypertensive drugs. Glomerular filtration rate (GFR), determined by Tc^{99m}DTPA renal clearance, and urinary protein excretion were monitored at 3-month intervals. Enalapril, in contrast to placebo therapy, was associated with an initial (40%) and sustained (33%) decrease in urinary protein excretion. Patients randomized to both enalapril or placebo experienced mean decreases in GFR, from 1.01 mL/s/1.73 m^2 (61 mL/min/1.73 m^2) to 0.85 mL/s/1.73 m^2 (51 mL/min/1.73 m^2), and from 1.06 mL/s/1.73 m^2 (64 mL/min/1.73 m^2) to 0.97 mL/s/1.73 m^2 (58 mL/min/1.73 m^2), respectively. Eleven of 18 patients (61%) randomized to enalapril, and 10 of 15 (66%) patients randomized to placebo, had a decrease in GFR; their rates of progression were −1.18 mL/min/1.73 m^2/mo and −1.00 mL/min/1.73 m^2/mo, respectively. In the absence of changes in blood pressure, the addition of an ACE inhibitor to patients with clinical diabetic nephropathy could not be shown to confer a unique renal protective effect. A prolonged decrease in 24-hour protein excretion could not be shown to predict attenuation in the progression of established clinical diabetic nephropathy.
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INDEX WORDS: Diabetes mellitus; proteinuria; glomerular filtration rate; angiotensin-converting enzyme inhibition.

DIABETIC NEPHROPATHY is the most common cause of end-stage renal disease in the United States.[1] Potential therapeutic modalities that may prevent the development, or attenuate the progression, of diabetic nephropathy include (1) strict metabolic control, (2) strict control of systemic hypertension, (3) strict control of "glomerular hypertension", and/or (4) dietary protein restriction.

The stage of diabetic nephropathy termed "overt" or "clinical" diabetic nephropathy is characterized by persistent proteinuria (urinary protein excretion > 500 mg/24 h), often systemic hypertension, and a constant and progressive decline in glomerular filtration rate (GFR).[2-4] Uncontrolled systemic hypertension is a powerful accelerating factor in the course of clinical diabetic nephropathy.[2-4]

Angiotensin-converting enzyme (ACE) inhibitors are effective drugs in controlling the systemic hypertension accompanying clinical diabetic nephropathy.[5-12] Furthermore, they have been demonstrated to be antiproteinuric,[5,8-13] a pharmacological property that has been equated by some investigators to reflect attenuation in the rate of progression of clinical diabetic nephropathy. However, there are no reports of experimental animal studies, or randomized, double-blind, placebo-controlled clinical trials, that have tested the hypothesis that a reduction in established proteinuria does indeed reflect attenuation in the rate of progression of clinical diabetic nephropathy, or the hypothesis that ACE inhibitor therapy can attenuate the rate of progression of established clinical diabetic nephropathy independent of strict control of systemic hypertension.

We report the results of a randomized, double-blind, clinical trial designed to compare the longitudinal (18-month) effect of an ACE inhibitor (enalapril) and a placebo on urinary protein excretion and the rate of progression of renal disease, in 33 patients with clinical diabetic nephropathy. Systemic blood pressure was controlled with the use of conventional antihypertensive drugs. Our results confirm that ACE inhibitor therapy is antiproteinuric, independent

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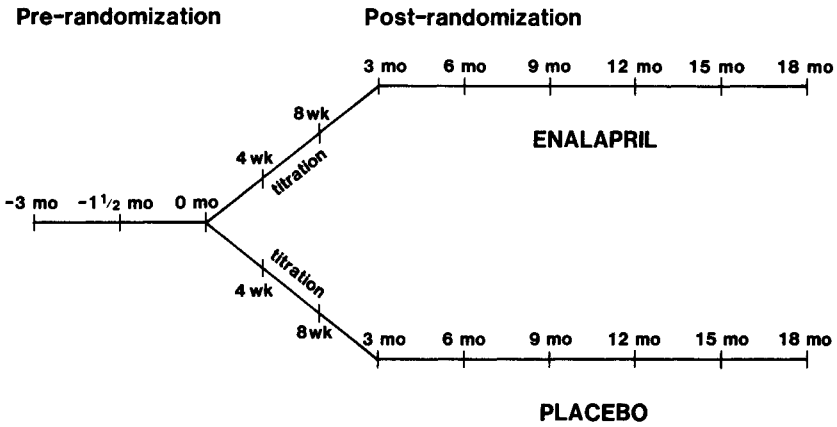

Fig. 1. Protocol: Prerandomization phase: patients evaluated at −3 months, −1.5 months, and 0 month; postrandomization phase: titration of drug therapy followed by evaluation at 3-month intervals until completion of the trial (18 months).

of systemic blood pressure control. However, the rates of progression of renal disease were not significantly different between the two treatment groups. Furthermore, a prolonged decrease in 24-hour urinary protein excretion did not predict attenuation of progression of established disease.

METHODS

Selection of Patients

Patients were selected and randomized for study if they had (1) a history of either type I or type II diabetes mellitus; (2) clinical diabetic nephropathy defined by the presence of persistent proteinuria greater than 500 mg/24 h before drug randomization; (3) no clinical or laboratory evidence of primary renal disease; (4) creatinine clearance greater than 20 mL/min/1.73 m^2; (5) clinical evidence of additional diabetic target organ damage (diabetic retinopathy and/or peripheral neuropathy); and (6) persistent diastolic blood pressures less than 90 mm Hg on either no drug therapy or on conventional antihypertensive drugs excluding ACE inhibitors and calcium antagonists. Patients were excluded from the study if they had (1) a glycosylated hemoglobin (HbA_{1c}) consistently greater than 12% (average blood glucose > 16.7 mmol/L [300 mg/dL]); (2) noncompliance with drug therapy; (3) a history of hyperkalemia or baseline serum potassium greater than 5.2 mmol/L; (4) decompensated congestive heart failure; (5) unstable angina pectoris or history of a myocardial infarction within 6 months of entering the protocol; (6) cerebrovascular disease within 12 months of entering the protocol; (7) females with the potential to become pregnant; (8) a history of hypersensitivity reaction to ACE inhibitor peptide-like drugs; and/or (9) evidence of clinically important bladder dysfunction, negating the ability to replicate renal clearance determinations. The clinical trial was approved by the University of Missouri Health Sciences Institutional Review Board; all patients gave written informed consent before enrollment into the study.

Fifty patients were enrolled in the study. Of these, 42 patients completed the 3-month prerandomization stabilization period (see below). Of these 42 patients, 33 met all of the inclusion criteria and completed a minimum 3-month prospective follow-up evaluation. They are the subject of this report. Of the nine patients excluded from the postrandomization analysis, three had a 24-hour urinary protein excretion less than 500 mg just prior to randomization (0 month), one had a creatinine clearance less than 20 mL/min/1.73 m^2 at 0 month, and three had uncontrolled systemic blood pressure (ie, diastolic blood pressure > 90 mm Hg) on conventional antihypertensive drug therapy at 0 month. Two patients randomized to enalapril did not complete more than 4 weeks of the protocol; one patient was noncompliant and one patient died (sudden death, not believed to be related to drug therapy).

Prerandomization Stabilization Period

Because of the potential for biological and laboratory variability in the parameters to be assessed (eg, dietary protein intake, HbA_{1c}, systemic blood pressure, renal function indices), all patients selected for study underwent a 3-month run-in period before placement on either placebo or enalapril therapy. During this prerandomization period, patients underwent assessment of all study parameters (see below) at −3 months, at −1.5 months, and at 0 month (Fig 1). The mean of these three determinations for each parameter was considered the baseline value for each parameter. Throughout the protocol, patients were advised to make no change in dietary protein or sodium intake. Diabetic and blood pressure control were monitored and treated by the patient's individual physician.

Study Parameters and Methods

Because of the potential influence of dietary protein intake on the rate of progression of renal disease, all patients underwent assessment of dietary protein intake as estimated from 24-hour urine urea nitrogen (UUN) excretion (estimated dietary protein intake = $[(UUN)_{24hr} \times 6.25]/0.80$).[14] Diabetic control was assessed by monitoring HbA_{1c} determined by high-performance liquid chromatography.[15] Systemic blood pressure was assessed in three positions: (1) 5-minute recumbent, (2) 5-minute sitting, and (3) 2-minute upright. All determinations were made in triplicate, using a standard mercury manometer, to record systolic (first-phase) and diastolic (fifth-phase) blood pressure. The three pressures in each position were averaged to give a mean representation of the patient's blood pressure.

Serum and 24-hour urine were collected for determinations of sodium, potassium, creatinine, and urea nitrogen. Electrolytes were determined by ion-selective electrode; urea and creatinine were determined by standard spectrophotometric methods. Urinary protein was determined by the Coomassie brilliant blue binding method of Bradford.[16] The 24-hour urinary excretion of electrolytes and protein were expressed as millimoles or grams per gram creatinine, respectively, to normalize for accuracy of urine collection. Plasma was collected for the measurement of recumbent plasma renin activity (PRA) and assayed as previously described.[17]

Timed renal clearances were performed as previously described.[17,18] Patients were hydrated (20 mL tap water/kg body weight) to achieve a urine specific gravity less than 1.010. GFR was determined initially by three different methods: clearances of creatinine, inulin, and Tc^{99m}DTPA. Tc^{99m}DTPA clearance was determined using a 2-μCi single intravenous injection technique.[19] Effective renal plasma flow (ERPF) was determined by *p*-aminohippurate (PAH) clearance. Priming doses of inulin and PAH were administered to obtain a plasma inulin concentration of approximately 20 mg/dL, and a plasma PAH concentration of approximately 2 mg/dL. Subsequently, a sustaining infusion of inulin and PAH (rate depending on the estimated level of GFR and ERPF) was administered to maintain the initial plasma concentrations of inulin and PAH. Three timed urine collections (spontaneous voidings) were made at 25- to 40-minute intervals following a 75-minute equilibration period. Urine for determination of PAH was alkalinized to prevent loss of free PAH in glycosuric urine.[20] Plasma was obtained before infusion of Tc^{99m}DTPA, inulin, and PAH, and after the bolus infusions of Tc^{99m}DTPA, inulin, and PAH, between each urine collection. The specific activity of Tc^{99m}DTPA was assayed by gamma counter. Plasma and urine were assayed for inulin and PAH as previously described.[17,18] Clearances of Tc^{99m}DTPA, inulin, PAH, and creatinine were calculated as (urine concentration/plasma concentration) $\times$ urine flow rate (mL/min) and corrected for body surface area (1.73 m^2). The initial body surface area was used for all subsequent clearance calculations. All results were expressed as single mean values of triplicate determinations for each patient. Filtration fraction was calculated as the clearance of Tc^{99m}DTPA divided by clearance of PAH, and expressed as a percent of the clearance of PAH.

Thirty-one patients undergoing simultaneous determinations of creatinine, inulin, and Tc^{99m}DTPA clearances at 0 month demonstrated (mean $\pm$ SEM) a creatinine clearance of 1.32 $\pm$ 0.10 mL/s/1.73 m^2 (79 $\pm$ 6 mL/min/1.73 m^2), an inulin clearance of 1.10 $\pm$ 0.12 mL/s/1.73 m^2 (66 $\pm$ 7 mL/min/1.73 m^2), and a Tc^{99m}DTPA clearance of 1.05 $\pm$ 0.08 mL/s/1.73 m^2 (63 $\pm$ 5 mL/min/1.73 m^2). Creatinine clearance was significantly greater ($P < 0.001$) than either the inulin or Tc^{99m}DTPA clearances; inulin and Tc^{99m}DTPA clearances were not statistically different from each other, and were significantly correlated ($r = 0.9091$, $P < 0.0001$). Because the serum creatinine and the creatinine clearance may not accurately reflect changes in GFR,[18,19] and because there was both a worldwide shortage of inulin at the time the protocol was initiated and concern about the potential effect of fluctuating plasma glucose concentrations on plasma and urinary determinations of inulin during the 4 hours required to conduct a renal clearance study, Tc^{99m}DTPA renal clearances were performed in all patients at most points throughout the protocol, and these results were used to assess serial change in GFR.

Postrandomization Protocol

The baseline characteristics of the 33 patients entering into the postrandomization phase of the protocol are given in Table 1. The groups were similar with regard to age, sex, race, weight, body surface area, diabetic target organ damage, and hypertension. The primary difference between the placebo and enalapril treatment groups was the absence of type I diabetic patients randomized to placebo therapy.

Following randomization, patients were titrated on drug therapy at 2-week intervals. The initial dose was one 5-mg tablet of placebo or enalapril (both drugs provided by Merck, Sharp & Dohme Research Laboratories, West Point, PA). Subsequently, the patients were titrated to one 10-mg tablet of placebo or enalapril, one 20-mg tablet of placebo or enalapril, or two 20-mg tablets of placebo or enalapril. The maximum dose prescribed was that which did not clinically lower baseline sitting blood pressure, or clinically produce side effects (eg, dizziness, weakness, orthostatic hypotension). Drug dose was back titrated to a prior drug dose if the patient was intolerant to the higher dose. Compliance was assessed by count of returned tablets. The number of patients on each drug dose level, at their last completed clinic visit, are given in Table 2. Concurrent conventional antihypertensive drug therapy prescribed at the same time is also given in Table 2.

Following baseline assessment of the study parameters, and randomization to drug therapy, patients underwent assessment of the study parameters at 3-month intervals for a total of 18 months. Of the 15 patients randomized to placebo therapy, two patients completed only 3 months of therapy; subsequently, one patient moved and one patient experienced a cerebrovascular accident. One of the 15 patients randomized to placebo therapy was diagnosed to have small cell carcinoma of the lung following 9 months of therapy. Thus, 12 of the 15 placebo patients completed 18 months of therapy. Of the 18 patients randomized to enalapril therapy, two patients completed 6 months of therapy; subsequently, one patient moved and one patient died following an elective surgical procedure. Three of the remaining 16 patients randomized to enalapril therapy did not complete the protocol: one patient suffered an acute myocardial infarction following 9 months of therapy, one patient moved after 12 months of therapy, and one patient was begun on hemodialysis therapy after 15 months of therapy. Thus, 13 of the 18 enalapril patients completed 18 months of therapy.

Statistical Analysis

All results are given as a mean $\pm$ SEM. To assess prerandomization stability of the study parameters, Friedman's test was applied to each of the variables to determine whether the three values (−3 months, −1.5 months, and 0 month) differed. Where differences occurred, pairwise comparisons were made (Wilcoxon signed-rank test).[21]

Since, with rare exception (see Results), there were no statistical differences between the three prerandomization values, these three values were averaged for each study parameter to give a baseline value. The two treatment groups were then

Table 1. Baseline Patient Characteristics

	Placebo	Enalapril
Physical characteristics		
No. of subjects	15	18
Age (yr) (range)	57.0 (47-70)	44.1 (26-66)
Sex	13 M, 2 F	11 M, 7 F
Race	14 C, 1 B	16 C, 2 B
Weight (kg) (range)	93.2 (60.1-125.5)	83.8 (55.5-121.1)
Surface area (m^2) (range)	2.07 (1.64-2.56)	1.95 (1.44-2.30)
Diabetes mellitus		
Type I/type II	0/15	10/8
Duration (yr) (range)	14.1 (1-35)	20.4 (7-42)
Treatment: insulin	10	18
Treatment: oral hypoglycemics	5	0
Control ($HbA_1C \leq 10\%$)	13 of 15 (87%)	11 of 18 (61%)
Target organ damage		
Nephropathy	15	18
Retinopathy	10	18
Neuropathy	13	14
Hypertension		
History and drug treatment	10/15	12/18
Duration (yr) (range)	12.5 (1-41)	6.4 (1-15)
Specific antihypertensive drugs	Atenolol, prazosin	Furosemide, atenolol
	Hydrochlorothiazide, atenolol	Furosemide, atenolol, minoxidil
	Metolazone, atenolol, minoxidil	Furosemide
	Furosemide, metolazone	Furosemide, prazosin
	Furosemide	Hydrochlorothiazide, hydralazine
	Hydrochlorothiazide, prazosin	Prazosin
	Furosemide	Furosemide
	Furosemide	Hydrochlorothiazide, propranolol
	Hydrochlorothiazide	Hydrochlorothiazide
	Maxzide	Atenolol
		Hydrochlorothiazide
		Furosemide

compared using these baseline values for each study parameter (Wilcoxon rank-sum test).[21]

The experimental design for this study was a standard two-factor (treatment, time) design with repeated measures on one factor (time). Several different statistical methods have been proposed for use with this design. The classical analysis of variance approach first looks at the interaction term. This requires normally distributed data and does not make good use of data obtained from subjects who do not complete the study. We chose to employ statistical methods that allow us to use data from a subject for as long as the subject remained in the study. A specific example of one such approach is that proposed by Laird and Ware,[22] which allows for subjects to have different numbers of observations over time and also allows for a linear function to represent the progress of a subject over time. However, this approach requires the assumption of normally distributed data. We employed statistical methods that allowed us to use data from all subjects for as long as they remained in the study, but did not require the assumption of normality. To determine if the effects of the treatments were different, we employed methods similar to those of Knoke,[23] Crager,[24] and Samuels.[25] Specifically, the groups were compared at each time period by employing a nonparametric analysis of covariance using the baseline value as a covariate. This approach allows data to be used from a subject for as long as the subject is in the study. However, at the last time period, only completers were used in the analysis.

Another approach to determine if treatment effects were different is to see if the rate of change for an outcome variable is different between the two groups. The rate of change on a given study parameter was determined by either obtaining the difference between the last protocol visit and baseline value and dividing by the total treatment time (ie, calculated rate of change), or by fitting a regression line to the data from each subject (ie, linear regression analysis). In either case, each subject's slope was used as the data point to compare the rates of change between the two groups, using the Wilcoxon rank-sum test. We also compared the two groups using the slope data by again employing an analysis of covariance with the baseline and time in study as covariates. In addition, 95% confidence limits for the mean rates of decrease in GFR were calculated as: Mean $\pm$ $t_{.975}$ (n $-$ 1)SEM, where $t_{.975}$ (n $-$ 1) is the 97.5 percentile of Student's *t* distribution with n $-$ 1 degrees of freedom, SEM is the standard error of the mean, and n is the number of subjects in a particular group.

Within each group, we compared the outcome variables at each time period to the baseline value using the Wilcoxon signed-rank test. This also allowed us to use all of the data

Table 2. Drug Therapy at Last Completed Clinic Visit

	Placebo	Enalapril
Experimental drug (N = number of patients)		
Dosage A	N = 1	N = 4 (5 mg/d)
Dosage B	N = 3	N = 2 (10 mg/d)
Dosage C	N = 3	N = 2 (20 mg/d)
Dosage D	N = 8	N = 10 (40 mg/d)
Concurrent antihypertensive drug therapy (total dose)		
	Hydrochlorothiazide (50 mg), prazosin (4 mg)	Furosemide (40 mg), atenolol (50 mg)
	Hydrochlorothiazide (100 mg)	Furosemide (80 mg), atenolol (50 mg), amiloride (5 mg)
	Metolazone (2.5 mg), atenolol (50 mg), minoxidil (10 mg)	Hydrochlorothiazide (25 mg), amiloride (5 mg)
	Metolazone (2.5 mg), furosemide (40 mg)	Furosemide (20 mg)
	Furosemide (80 mg)	Hydrochlorothiazide (50 mg), hydralazine (25 mg)
	Prazosin (3 mg)	Furosemide (40 mg)
	Furosemide (40 mg), atenolol (50 mg)	Hydrochlorothiazide (50 mg), propranolol (40 mg)
	Furosemide (20 mg), atenolol (50 mg)	Prazosin (1 mg)
	Furosemide (40 mg)	Dyazide (1/d), metolazone (2.5 mg)
	Furosemide (40 mg), prazosin (3 mg)	
	Maxzide (1/d), atenolol (50 mg)	

from a given subject for as long as the subject remained in the study. It is worth noting that when the 18-month values were compared with baseline values, only those subjects who completed the protocol were used.

Twenty-one of the 33 patients entering into the protocol demonstrated a decrease in GFR (as determined by Tc^{99m}DTPA renal clearance) at their last protocol visit. Since these patients did not represent a random sample from diabetic patients who would ultimately experience a progressive decrease in GFR over time, their postrandomization treatment responses to either placebo (N = 10) or enalapril (N = 11) are presented as descriptive data (mean ± SEM).

Finally, Spearman correlations were computed for the variables, as well as for change in the variables from baseline.[21]

In assessing differences both within and between groups, a statistical difference was considered present when $P < 0.05$; however, when multiple comparisons (tests), usually six of them, were made to answer a specific question, a statistical difference was considered present when $P < 0.01$ (approximate Bonferroni adjustment).

RESULTS

Prerandomization Stability and Baseline Values

Table 3 gives results for the assessment of stability of study parameters at −3 months, −1.5 months, and 0 month, and the baseline values for patients randomized to either placebo or enalapril therapy. The only significant differences between prerandomization values within a treatment group were placebo recumbent diastolic blood pressure and 24-hour sodium excretion, and enalapril serum potassium. The only significant differences in baseline values between the two treatment groups were HbA_{1c}, recumbent diastolic blood pressure, sitting systolic and diastolic blood pressures, and upright systolic blood pressure.

Postrandomization Treatment Effect

Estimated dietary protein intake was not significantly different from baseline throughout the protocol in either treatment group (mean ± SEM slopes: placebo, −0.01 ± 0.01; enalapril, −0.01 ± 0.01), nor were there significant differences between the two treatment group slopes. Similarly, dietary sodium intake was not significantly different from baseline throughout the protocol in either treatment group as estimated from 24-hour urinary sodium excretion (mean ± SEM slopes: placebo, −3.78 ± 3.40; enalapril, −1.85 ± 1.23), nor were there significant differences between the two treatment group slopes.

Although average baseline blood glucose, as assessed by HbA_{1c}, was significantly different between the two treatment groups (see Table 3), there was no significant change in HbA_{1c} from

Table 3. Prerandomization Data Assessment of Stability and Baseline Values

	−3 Months	−1.5 Months	0 Month	Baseline
HbA_1C (%)				
Placebo	8.6 ± 0.4	8.3 ± 0.3	8.2 ± 0.5 (n = 14)	8.4 ± 0.3
Enalapril	9.6 ± 0.5	9.5 ± 0.4	9.7 ± 0.4 (n = 17)	9.5 ± 0.4*
Recumbent BP (mm Hg)				
Placebo	156 ± 6/89 ± 2	156 ± 7/87 ± 2	152 ± 6/84 ± 2‡	155 ± 6/86 ± 2
Enalapril	141 ± 6/81 ± 2 (n = 17)	138 ± 5/80 ± 1	139 ± 5/79 ± 2	139 ± 5/80 ± 2†
Sitting BP (mm Hg)				
Placebo	154 ± 6/86 ± 2	151 ± 6/87 ± 3	146 ± 4/84 ± 1	150 ± 5/86 ± 2
Enalapril	139 ± 5/79 ± 3	136 ± 5/79 ± 2	133 ± 5/78 ± 2	136 ± 4*/79 ± 2*
Upright BP (mm Hg)				
Placebo	137 ± 7/79 ± 2 (n = 14)	139 ± 7/82 ± 3	132 ± 6/77 ± 3	136 ± 6/80 ± 2
Enalapril	123 ± 5/76 ± 3 (n = 17)	124 ± 4/75 ± 2	122 ± 5/72 ± 2	123 ± 4*/74 ± 2
Estimated protein intake (g/kg body wt)				
Placebo	0.93 ± 0.09	0.88 ± 0.09	0.82 ± 0.09	0.88 ± 0.07
Enalapril	0.91 ± 0.08	0.88 ± 0.07	0.79 ± 0.13 (n = 16)	0.87 ± 0.07
24-h sodium excretion (mmol/g creatinine)				
Placebo	147 ± 15	137 ± 18	122 ± 16‡	135 ± 13
Enalapril	148 ± 10	164 ± 15	148 ± 20 (n = 16)	154 ± 10
24-hr creatinine excretion (g/total vol)				
Placebo	1.44 ± 0.13	1.46 ± 0.12	1.46 ± 0.13	1.45 ± 0.11
Enalapril	1.23 ± 0.11	1.13 ± 0.09	1.23 ± 0.13 (n = 16)	1.19 ± 0.09
24-hr protein excretion (g/g creatinine)				
Placebo	2.32 ± 0.45	1.98 ± 0.45	2.00 ± 0.44	2.10 ± 0.42
Enalapril	2.03 ± 0.33	2.18 ± 0.38	2.13 ± 0.40 (n = 16)	2.08 ± 0.32
Creatinine clearance (mL/s/1.73 m^2)				
Placebo	1.42 ± 0.18	1.38 ± 0.15 (n = 14)	1.40 ± 0.15	1.40 ± 0.15
Enalapril	1.18 ± 0.13	1.22 ± 0.15 (n = 17)	1.17 ± 0.12	1.18 ± 0.12
Tc^{99m}DTPA clearance (mL/s/1.73 m^2)				
Placebo	1.13 ± 0.13 (n = 14)	1.00 ± 0.10	1.08 ± 0.13	1.07 ± 0.12
Enalapril	0.95 ± 0.10 (n = 15)	1.00 ± 0.10 (n = 16)	0.97 ± 0.10	1.02 ± 0.10
Inulin clearance (mL/s/1.73 m^2)				
Placebo	1.09 ± 0.17 (n = 4)	1.33 ± 0.68 (n = 2)	1.16 ± 0.18 (n = 13)	1.13 ± 0.17 (n = 14)
Enalapril	0.89 ± 0.17 (n = 6)	0.95 ± 0.27 (n = 3)	1.06 ± 0.14 (n = 16)	1.03 ± 0.13 (n = 16)

Table 3. Prerandomization Data Assessment of Stability and Baseline Values (cont'd)

	−3 Months	−1.5 Months	0 Month	Baseline
PAH clearance (mL/s/1.73 m^2)				
Placebo	4.93 ± 0.62	5.07 ± 0.48 (n = 14)	4.77 ± 0.58	4.85 ± 0.53
Enalapril	5.05 ± 0.60	4.78 ± 0.50 (n = 17)	4.27 ± 0.53 (n = 17)	4.82 ± 0.52
PRA (ng/[L · s])				
Placebo	0.8 ± 0.2	0.9 ± 0.2 (n = 14)	0.9 ± 0.2	0.9 ± 0.1
Enalapril	1.2 ± 0.4	1.3 ± 0.4 (n = 16)	1.0 ± 0.4 (n = 16)	1.1 ± 0.3
Serum potassium (mmol/L)				
Placebo	4.0 ± 0.1	4.0 ± 0.1	4.0 ± 0.1	4.0 ± 0.1
Enalapril	4.2 ± 0.1	4.1 ± 0.1	4.3 ± 0.1§	4.2 ± 0.1

NOTE. Placebo: N = 15, enalapril: N = 18. Values are mean ± SEM.

Between-group statistical comparison, baseline values: *$P < 0.05$ compared with placebo, †$P < 0.02$ compared with placebo.

Within-group statistical comparison, values at −3, −1.5, and 0 months: ‡$P < 0.05$ 0 month compared with −3 months, §$P < 0.05$ 0 month compared with −1.5 months.

baseline values throughout the protocol in either treatment group (mean ± SEM slopes: placebo, −0.08 ± 0.07; enalapril, −0.02 ± 0.02), nor were there significant differences in HbA_{1c} between the two treatment group slopes.

Although there were differences in some baseline systemic blood pressure values between the two treatment groups (see Table 3), systemic blood pressure values were not significantly different from baseline values (using slope data) throughout the protocol in either treatment group (see Fig 2 [placebo] and Fig 3 [enalapril]). In general, patients randomized to enalapril therapy had lower mean baseline recumbent, sitting, and upright systemic blood pressures values. However, there were no significant differences in systemic blood pressure values (using slope data) between the two treatment groups.

Baseline PRA was not significantly different between the two treatment groups (see Table 3). Enalapril therapy, in contrast to placebo therapy, significantly increased PRA during the study (Fig 4).

Baseline 24-hour urinary protein excretion was not significantly different between the two treatment groups (3.0 g/24 h placebo *v* 2.5 g/24 h enalapril) (see Table 3). Enalapril therapy, in contrast to placebo therapy, significantly decreased urinary protein excretion during the study (Fig 5 and Table 4). Fourteen of the 18 patients randomized to enalapril therapy exhibited a sustained decrease (−51%) in urinary protein excretion (from 2.40 g/g creatinine to 1.17 g/g creatinine at the last protocol visit). Changes in 24-hour urinary protein excretion (baseline to 3 months, and baseline to last protocol visit) were not significantly correlated with changes in systemic blood pressure, $Tc^{99m}DTPA$ clearance, PAH clearance, filtration fraction, or estimated dietary protein intake.

Patients randomized to either placebo or enalapril therapy demonstrated a decrease in GFR as assessed by $Tc^{99m}DTPA$ clearance (Fig 6 and Table 4). In patients randomized to placebo therapy, baseline GFR was 1.06 mL/s/1.73 m^2 (64 mL/min/1.73 m^2), and decreased to 0.97 mL/s/1.73 m^2 (58 mL/min/1.73 m^2) at the last protocol visit, over a mean 15.4-month treatment period. In patients randomized to enalapril therapy, baseline GFR was 1.01 mL/s/1.73 m^2 (61 mL/min/1.73 m^2), and decreased to 0.85 mL/s/1.73 m^2 (51 mL/min/1.73 m^2) at the last protocol visit, over a mean 15.7-month treatment period.

Ten of the 15 patients randomized to placebo therapy (66%), and 11 of the 18 patients randomized to enalapril therapy (61%), demonstrated a

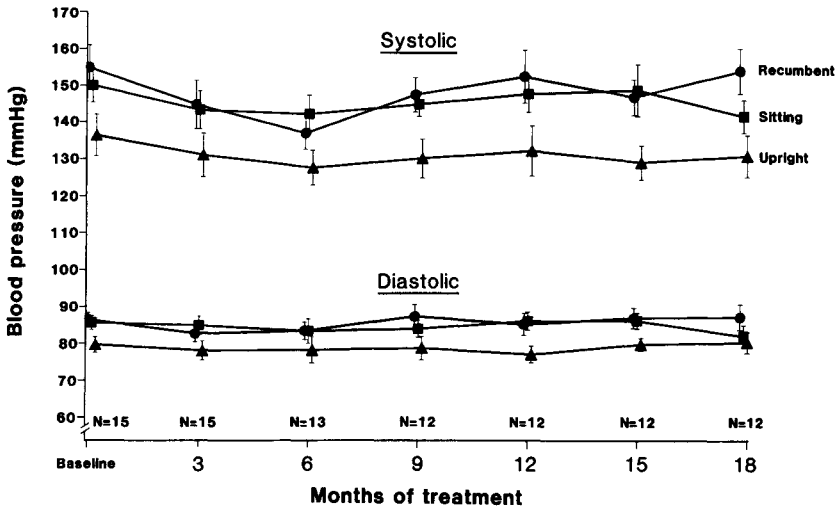

Fig. 2. Mean ± SEM systemic blood pressure response: placebo, N = 15. Mean ± SEM blood pressure slopes: recumbent blood pressure (●), systolic 0.47 ± 0.49, diastolic 0.25 ± 0.34; sitting blood pressure (■), systolic 0.17 ± 0.54, diastolic −0.07 ± 0.42; upright blood pressure (▲), systolic −0.13 ± 0.38, diastolic 0.19 ± 0.25.

decrease in GFR at their last protocol visit as assessed by Tc^{99m}DTPA clearance (Fig 7 and Table 4). In the patients randomized to placebo therapy, baseline GFR was 0.93 mL/s/1.73 m^2 (56 mL/min/1.73 m^2), and decreased to 0.73 mL/s/1.73 m^2 (44 mL/min/1.73 m^2). In the patients randomized to enalapril therapy, baseline GFR was 1.15 mL/s/1.73 m^2 (69 mL/min/1.73 m^2), and decreased to 0.83 mL/s/1.73 m^2 (50 mL/min/1.73 m^2).

In patients receiving placebo therapy, the rate of decrease in GFR (calculated and by linear regression analysis) was less during the first three months of therapy compared to the remaining months of therapy (see Table 4). In patients receiving enalapril therapy, the rate of decrease in GFR (calculated and by linear regression analysis) was greater during the first 3 months of therapy compared to the remaining months of therapy (see Table 4). There were no statistical differences in the rates of progression of renal disease, as determined by either calculated or linear regression analysis, between the two treatment groups.

Although all of the patients randomized to placebo therapy had type II diabetes mellitus, only 10 of the 18 patients randomized to enalapril therapy had type II diabetes mellitus. Table 5 compares the baseline characteristics and renal function responses of the type I and type II diabetic patients randomized to enalapril therapy. There were no significant differences in treatment times, baseline HbA_{1c}, recumbent systemic blood pressure, estimated protein intake, PRA, 24-hour urinary sodium and creatinine excretion, GFR

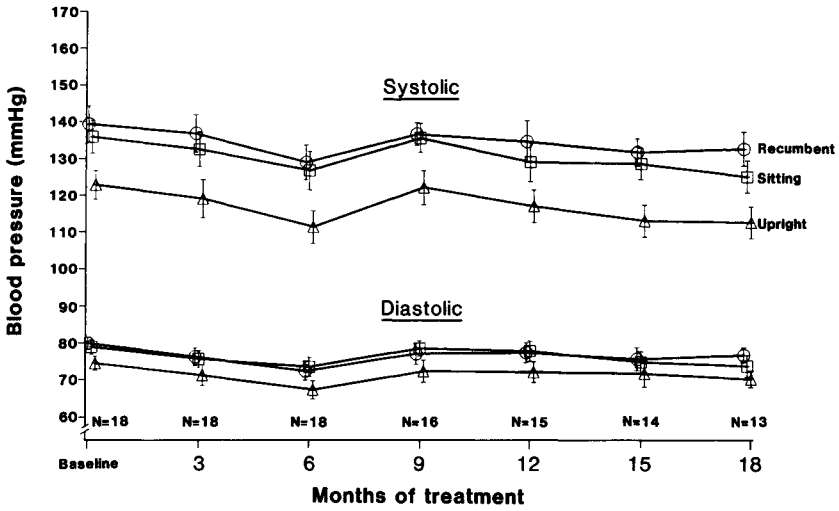

Fig. 3. Mean ± SEM systemic blood pressure response: enalapril, N = 18. Mean ± SEM blood pressure slopes: recumbent blood pressure (○), systolic −0.99 ± 0.52, diastolic −0.69 ± 0.45; sitting blood pressure (□), systolic −1.57 ± 0.99, diastolic −0.32 ± 0.35; upright blood pressure (△), systolic −0.18 ± 0.33, diastolic 0.06 ± 0.22.

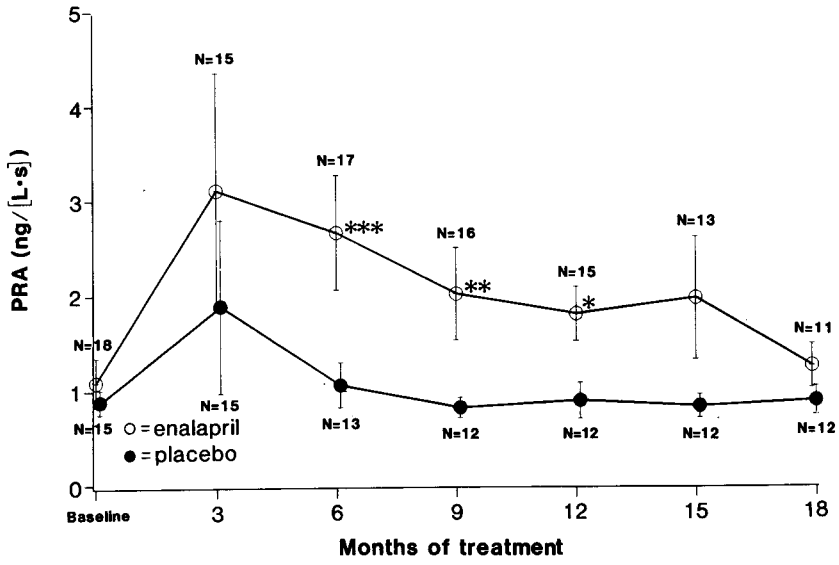

Fig. 4. Plasma renin activity response: effect of placebo (●) and enalapril (○) on PRA. *$P < 0.05$ compared with baseline; **$P < 0.02$ compared with baseline; ***$P < 0.01$ compared with baseline.

as assessed by Tc^{99m}DTPA, or 24-hour urinary protein excretion. Their calculated and linear regression analysis rates of decrease in GFR were not statistically different from each other. Thus, type I and type II diabetic patients with clinical diabetic nephropathy demonstrated similar renal responses to ACE inhibitor therapy.

Baseline ERPF was not significantly affected by either placebo or enalapril therapy (mean ± SEM slopes: placebo, −1.36 ± 1.52; enalapril, −1.17 ± 1.48) (see Table 4). Baseline filtration fraction was not significantly affected by either placebo or enalapril therapy, (mean ± SEM slopes: placebo, −0.001 ± 0.002; enalapril, −0.001 ± 0.001), although patients randomized to enalapril therapy did demonstrate a significant decrease in filtration fraction at their last protocol visit (see Table 4).

Serum potassium was not significantly different from baseline throughout the protocol in either treatment group (mean ± SEM potassium slopes: placebo, 0.02 ± 0.01; enalapril, 0.01 ± 0.01), nor were there significant differences between the two treatment group slopes. Only two patients on enalapril therapy exhibited evidence of clinically important hyperkalemia (serum po-

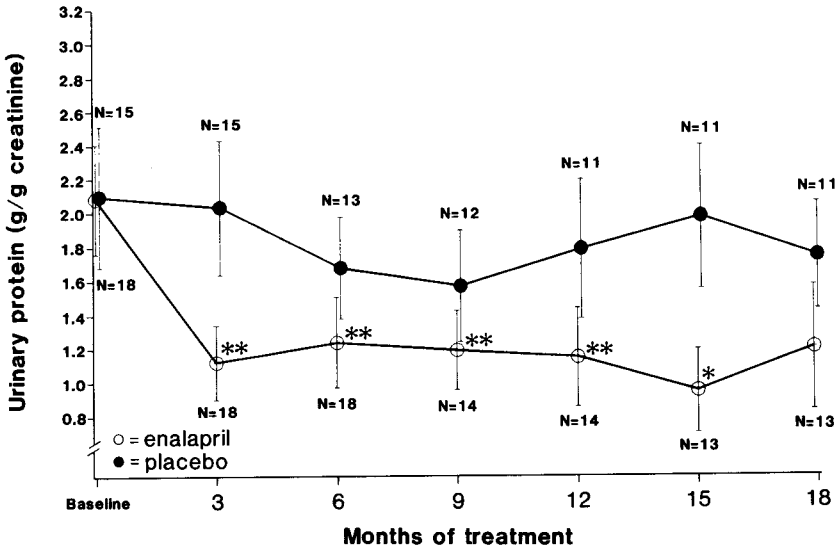

Fig. 5. Urinary protein excretion response: effect of placebo (●) and enalapril (○) on 24-hour urinary protein excretion. *$P < 0.02$ compared with baseline; **$P < 0.01$ compared with baseline.

Table 4. Initial Versus Late Treatment Effects on Renal Function

	Baseline	3 Months	Last Visit
24-hr urinary protein (g/g creatinine)			
Placebo			
All pts (N = 15)	2.10 ± 0.42	2.03 ± 0.40	1.77 ± 0.32 (n = 14)
Pts with ↓ GFR (N = 10)	2.45 ± 0.53	2.66 ± 0.47	2.26 ± 0.44 (n = 9)
Enalapril			
All pts (N = 18)	2.08 ± 0.32	1.12 ± 0.22§	1.40 ± 0.31†
Pts with ↓ GFR (N = 11)	2.36 ± 0.46	1.44 ± 0.32	1.70 ± 0.42
GFR (Tc^{99m}DTPA) (mL/s/1.73 m^2)			
Placebo			
All pts (N = 15)	1.06 ± 0.12	1.05 ± 0.12	0.97 ± 0.14*
Pts with ↓ GFR (N = 10)	0.93 ± 0.13	0.92 ± 0.15	0.73 ± 0.12
Enalapril			
All pts (N = 18)	1.01 ± 0.10	0.88 ± 0.10‡	0.85 ± 0.13‡
Pts with ↓ GFR (N = 11)	1.15 ± 0.13	0.98 ± 0.13	0.83 ± 0.13
ERPF (PAH) (mL/s/1.73 m^2)			
Placebo			
All pts (N = 15)	4.86 ± 0.53	4.77 ± 0.55	4.24 ± 0.58
Pts with ↓ GFR (N = 10)	4.50 ± 0.72	4.47 ± 0.80	3.43 ± 0.63
Enalapril			
All pts (N = 18)	4.81 ± 0.52	4.73 ± 0.60 (n = 16)	4.78 ± 0.58
Pts with ↓ GFR (N = 11)	5.55 ± 0.75	5.05 ± 0.85	5.00 ± 0.96
Filtration fraction (%)			
Placebo			
All pts (N = 15)	22.3 ± 1.2	22.7 ± 1.6	23.6 ± 1.5
Pts with ↓ GFR (N = 10)	21.7 ± 1.6	21.4 ± 1.8	22.9 ± 2.2
Enalapril			
All pts (N = 18)	21.5 ± 0.8	19.1 ± 1.5 (n = 16)	19.2 ± 1.2§
Pts with ↓ GFR (N = 11)	21.6 ± 1.1	20.9 ± 2.0	18.9 ± 1.6
	Base-3 mo	**3 mo-last visit**	**Base-last visit**
Calculated rate of ↓ GFR (mL/min/1.73 m^2/mo)			
Placebo			
All pts (N = 15)	−0.09 ± 0.77 (−1.74, 1.56)	−0.41 ± 0.29 (−1.04, 0.22) (n = 13)	−0.37 ± 0.29 (−0.99, 0.25)
Pts with ↓ GFR (N = 10)	−0.17 ± 0.52 (−1.35, 1.01) (n = 9)	−0.85 ± 0.20 (−1.31, −0.39)	−0.92 ± 0.26 (−1.51, −0.33)
Enalapril			
All pts (N = 18)	−2.61 ± 0.95 (−4.62, −0.60) (n = 17)	−1.00 ± 0.71 (−2.51, 0.51) (n = 17)	−0.81 ± 0.27 (−1.38, −0.24)
Pts with ↓ GFR (N = 11)	−3.30 ± 1.44 (−6.51, −0.09)	−1.95 ± 1.00 (−4.18, 0.28)	−1.49 ± 0.29 (−2.14, −0.84)

Table 4. Initial Versus Late Treatment Effects on Renal Function (cont'd)

	3 mo-last visit	Base-last visit
Linear regression rate of ↓ GFR (mL/min/1.73 m²/mo)		
Placebo		
All patients (N = 15)	−0.48 ± 0.28 (−1.09, 0.13) (n = 13)	−0.43 ± 0.30 (−1.07, 0.21)
Pts with ↓ GFR (N = 10)	−0.90 ± 0.19 (−1.34, −0.46) (n = 9)	−1.00 ± 0.25 (−1.57, −0.43)
Enalapril		
All pts (N = 18)	−0.89 ± 0.67 (−2.30, 0.52)	−0.58 ± 0.25 (−1.11, −0.05)
Pts with ↓ GFR (N = 11)	−1.87 ± 0.99 (−4.08, 0.34)	−1.18 ± 0.29 (−1.82, −0.53)

NOTE. Values are mean ± SEM; confidence intervals are provided in parentheses for rates of decrease (↓) of GFR.
* $0.05 < P < 0.10$ compared with baseline.
† $P < 0.05$ compared with baseline.
‡ $P < 0.02$ compared with baseline.
§ $P < 0.01$ compared with baseline.

tassium > 5.7 mmol/L); this occurred at 2 and at 15 months of therapy. Both episodes of hyperkalemia spontaneously reversed to normal at the subsequent protocol visit. No patient was discontinued from the protocol for a drug (placebo or enalapril) side effect.

DISCUSSION

Our double-blind, randomized, placebo-controlled clinical trial was designed to evaluate the longitudinal (18-month) effect of an ACE inhibitor on 24-hour urinary protein excretion and renal function in patients with clinical diabetic nephropathy, having controlled systemic blood pressure on conventional antihypertensive drugs. Based on previously reported clinical trials, it was hypothesized that ACE inhibitor therapy would be antiproteinuric and be renal protective (ie, attenuate the rate of progression of renal disease). Furthermore, it was hypothesized that a decrease in proteinuria would reflect renal protection.

Unlike previously described clinical trials in patients with clinical diabetic nephropathy, our study was designed to assess prerandomization

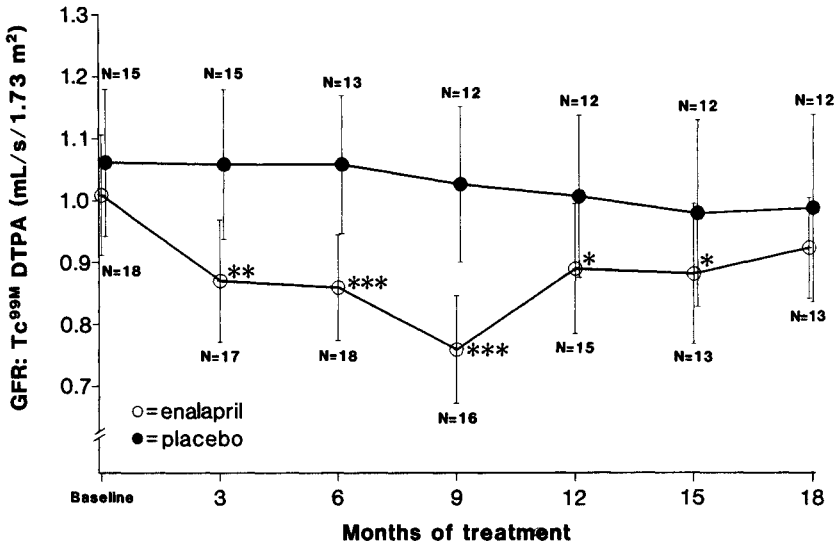

Fig. 6. GFR response: effect of placebo (●) and enalapril (○) on GFR as determined by Tc^{99m}DTPA renal clearance. *0.05 < P < 0.10 compared with baseline; **P < 0.02 compared with baseline; ***P < 0.01 compared with baseline.

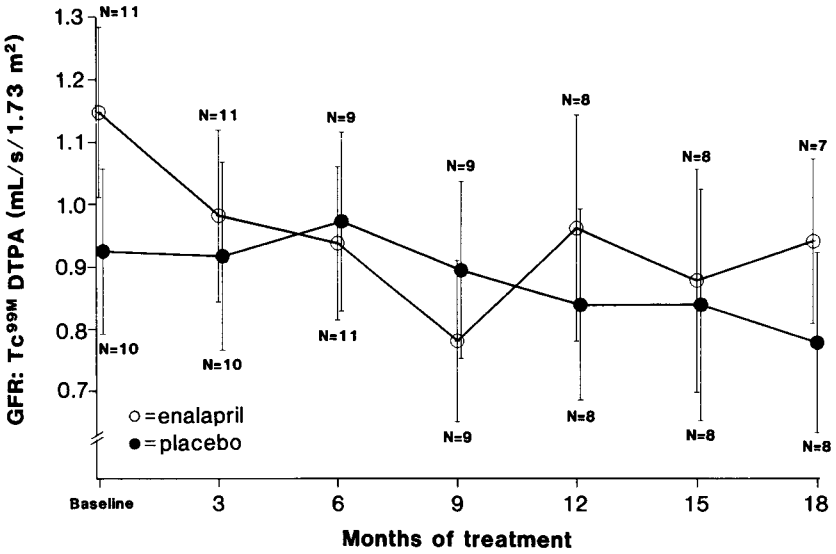

Fig. 7. GFR response: effect of placebo (●) and enalapril (○) on GFR in patients demonstrating a decrease in GFR at their last protocol visit, as determined by Tc^{99m}DTPA renal clearance.

stability of the biological parameters to be tested. With rare exception, there were no statistical differences in the prerandomization biological parameters tested. Therefore, we used the aggregate prerandomization data (−3 months, −1.5 months, and 0 month) to derive a representative 3-month baseline value for each biological parameter studied.

The results of our clinical trial demonstrate that those biological parameters which potentially could affect urinary protein excretion and renal function were not significantly altered from baseline by either placebo or enalapril therapy. Specifically, estimated dietary protein and sodium intake did not change throughout the protocol in either treatment group, nor were there differences between the two treatment groups. In general, patients consumed a diet containing 0.7 to 0.9 g/kg protein and 4 to 5 g of sodium. Although baseline HbA_{1c} values were higher in the patients randomized to enalapril therapy, postrandomization HbA_{1c} values were not different

Table 5. Type I Versus Type II Diabetic Patients Randomized to Enalapril Therapy

	Type I	Type II
No. of subjects	10	8
Treatment time (mo)	15.0 ± 1.6	16.5 ± 1.11
Baseline HbA_1C (%)	9.4 ± 0.5	9.7 ± 0.6
Baseline recumbent pressure (mm Hg)	133 ± 6/82 ± 3	147 ± 8/77 ± 2
Baseline estimated protein intake (g/kg body wt)	0.86 ± 0.11	0.87 ± 0.09
Baseline PRA (ng/[L · s])	1.1 ± 0.4	1.0 ± 0.4
Baseline 24-h sodium excretion (mmol/g creatinine)	165 ± 14	140 ± 12
Baseline 24-h creatinine excretion (g/total vol)	1.12 ± 0.11	1.28 ± 0.14
Baseline urinary protein excretion (g/g creatinine)	2.16 ± 0.47	1.99 ± 0.46
Urinary protein excretion at last protocol visit (g/g creatinine)	1.38 ± 0.36	1.42 ± 0.54
Baseline GFR (Tc^{99m}DTPA) (mL/s/1.73 m^2)	1.12 ± 0.15	0.88 ± 0.12
GFR at last protocol visit (mL/s/1.73 m^2)	0.97 ± 0.08	0.68 ± 0.15
Calculated rate decrease GFR (mL/min/1.73 m^2/mo)	−0.81 ± 0.42	−0.82 ± 0.35
	(−1.76, 0.14)	(−1.64, 0.01)
Linear regression rate decrease GFR (mL/min/1.73 m^2/mo)	−0.61 ± 0.40	−0.55 ± 0.31
	(−1.51, 0.29)	(−1.28, 0.18)

NOTE. Values are mean ± SEM; confidence intervals are provided in parentheses for rates of decrease GFR. There were no between-group statistical differences.

between the two treatment groups. Although some baseline systemic blood pressures were lower in the patients randomized to enalapril, neither drug therapy had a sustained directional affect on systemic blood pressure, and postrandomization systemic blood pressure slopes were not different between the two treatment groups.

As anticipated, enalapril therapy was associated with increases in PRA, whereas placebo therapy had no effect on PRA. The stimulation of PRA following ACE inhibitor therapy suggests good patient compliance with drug therapy; however, changes in concurrent antihypertensive drug therapy could have affected directional changes in PRA. Compliance, as judged by tablet count, was excellent throughout the protocol.

As anticipated, enalapril therapy was associated with a significant initial (46%) and sustained (33%) decrease in urinary protein excretion, whereas placebo therapy had no effect on urinary protein excretion. The well-known antiproteinuric effect of ACE inhibitor therapy has been attributed to (1) decrease in glomerular capillary hydraulic pressure, (2) increase in basement membrane barrier permselectivity, and/or (3) decrease in mesangial uptake and clearance of macromolecules.[3] Although our study did not directly address the mechanism(s) responsible for the antiproteinuric effect, our data confirm that the effect is unrelated to changes in systemic blood pressure;[13,26] furthermore, the antiproteinuric effect was unrelated to changes in GFR, ERPF, or filtration fraction.

Our results demonstrate that patients randomized to either placebo or enalapril therapy experienced a decrease in GFR. Specifically, 10 of 15 patients randomized to placebo therapy (66%), and 11 of 18 patients randomized to enalapril therapy (61%), experienced a decrease in GFR as assessed by Tc^{99m}DTPA renal clearances. In patients on placebo therapy, the predominant loss in GFR occurred after 3 months of therapy, whereas in patients on enalapril therapy there were decreases in GFR both before and following the first 3 months of therapy. The initial decrease in GFR associated with enalapril therapy may be attributed to a functional hemodynamic effect (ie, decrease in glomerular capillary hydraulic pressure); however, the filtration fraction (which may be an indirect marker of glomerular capillary hydraulic pressure) was not significantly decreased. Only at the last completed protocol visit was the filtration fraction significantly decreased in patients receiving enalapril therapy, and this decrease was the result of a decrease in GFR with preservation of ERPF.

The calculated and linear regression analysis rates of progression of renal disease were not different between the two treatment groups. This lack of a significant difference between the two treatment groups was present when analyzing all data points, from baseline to the last clinic visit, or when analyzing all data points beyond the initial 3 months of treatment to the last clinic visit. This latter analysis should mitigate the potential adverse influence of enalapril on GFR resulting from an initial decrease in glomerular capillary hydraulic pressure. At no time did patients randomized to placebo therapy experience a mean rate of deterioration in GFR greater than that experienced by patients randomized to enalapril therapy. Finally, the rates of progression of renal disease, observed by us in both treatment groups as determined by slope analysis, are consistent with the rates of progression of renal disease found in the published literature for type I diabetic patients with clinical diabetic nephropathy, treated with effective antihypertensive therapy.[8,27-34] To our knowledge, there are no similar prospective data published on type II diabetic patients with clinical diabetic nephropathy, treated with effective antihypertensive therapy.

Although all the patients randomized to placebo therapy had type II diabetes mellitus, only 10 of 18 patients randomized to enalapril therapy had type II diabetes mellitus. However, the clinical course of renal disease in the type I and type II diabetic patients randomized to enalapril therapy did not differ. Indeed, the calculated and linear regression analysis rates of progression of established diabetic nephropathy were not different between the type I and type II diabetic patients. Thus, there was no evidence that the relatively equal distribution of type I and type II diabetic patients to enalapril therapy, compared to the sole distribution of type II diabetic patients to placebo therapy, had any significant effect on renal function outcome in patients with established clinical diabetic nephropathy.

We conclude that patients with clinical diabetic nephropathy, having fair to good blood glucose control, sustained diastolic blood pressure less

than 90 mm Hg, and no change in dietary protein or sodium intake, could not be demonstrated to derive a unique renal protective effect from ACE inhibitor therapy. A potential criticism of our single-center clinical trial is the small sample size. If we were to consider our clinical trial as a pilot study, and used our observed variability in GFR to predict the sample size necessary to detect a statistically significant difference ($\alpha = 0.05$, power = 0.8) between the two treatment groups, it would require greater than 100 subjects per group to detect a difference in GFR of 0.12 mL/s/1.73 m^2 over 18 months, for a standard deviation of 0.55 mL/s/1.73 m^2. However, it required relatively few subjects to demonstrate that ACE inhibitor therapy does produce an initial and sustained decrease in 24-hour urinary protein excretion, unrelated to systemic or renal hemodynamic events. Taken together, our data clearly demonstrate that the antiproteinuric effect of ACE inhibition can not be assumed to predict attenuation of progression of established clinical diabetic nephropathy.

Our observations do not conflict with the existing experimental or clinical literature. There are no data published assessing the effect of ACE inhibitor therapy on the progression of established experimentally induced diabetic renal disease, and there are no clinical trials published that report a critical level of proteinuria below which clinical diabetic nephropathy is attenuated. Our observations in clinical diabetic nephropathy should not be extrapolated to diabetic patients with earlier phases of the disease (eg, microalbuminuria), where renal function and albumin excretion may both be normalized by ACE inhibitor therapy.[35-37] Furthermore, our observation should not be extrapolated to nondiabetic renal diseases, where ACE inhibitor therapy may control or arrest the primary pathophysiological event (eg, hypertensive nephrosclerosis).[38] Our observations do suggest that drug-mediated decreases in urinary protein excretion should not be interpreted as demonstrating "renal protection." Only carefully controlled randomized trials, using renal clearance techniques, can properly assess the potential ability of a drug or therapeutic maneuver to attenuate or arrest the rate of progression of renal disease.

ACKNOWLEDGMENT

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